

2(a)	$1.45 - 0.25 = 1.20$ $1.20 \div 0.25 = 4.8$ <i>allow ecf for mp2</i> $1.45 \div 0.25 = 5.8$ or $5.17 - 0.25 = 4.92$; $4.92 \div 0.25 = 20$; <i>allow ecf for mp2</i> $5.17 \div 0.25 = 21$	2
2(b)(i)	any two from: 1 different breeds at, same / named, location / environmental conditions, / have different milk yields ; 2 comparative data quote including units ; 3 (different breeds have) different gene pools / different alleles ; 4 due to selective breeding ;	2
2(b)(ii)	any two from: 1 same breed has different milk yields in, different location / named location / environmental conditions; 2 comparative data quote including units ; 3 how, climate / vegetation / veterinary care, affects milk yield ;	2

Question	Answer	Marks
2(c)	<i>any five from:</i> 1 human applies selection pressure / artificial selection ; 2 breed <u>Saanen</u> goat with, local / South African, goat ; 3 select offspring with, desired characteristics / high milk yield / adaptation to local conditions ; 4 breed offspring (with desired characteristics) ; 5 repeat (selection and breeding of offspring) for, several / many, generations ; 6 <i>ref. to</i> progeny testing / test milk yield in female offspring of male goats ; 7 avoid inbreeding / out-cross to unrelated goats ; 8 directional selection ; 9 AVP ; e.g. carry out programme in South Africa	5
2(d)	<i>any one advantage:</i> 1 stops / reduces, diarrhoea (in children) ; 2 improves children's, nutrition / growth / development ; <i>any one disadvantage:</i> 3 safety risks unknown / potential allergen / side effects ; 4 may disrupt ('good') bacteria / AW ; 5 expensive ; 6 not willing to drink GM milk ;	2